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Motherboard and CPU write-up

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My choice of motherboard is the **ASUS Prime B450M-A/CSM,** and my choice of CPU is the **AMD Ryzen 3 3200G**. The goal is to build a low-end computer for high school students. There are many reasons behind this exact choice of MB and CPU, but before even starting, it is a good idea to tell what the motherboard and the CPU are. The motherboard (MB) is the backbone of the computer; it is where all the parts are connected. The central processing Unit (CPU) is the brain of the computer; it handles all the processing tasks like executing instructions, manipulating data, and performing calculations. When building a computer, you must always look at the compatibility between all the components.

First, I chose the **ASUS Prime B450M-A/CSM** motherboard because its features are enough for students wanting a functioning and efficient device during their academic years. This Motherboard is from the ASUS brand. It has a Memory Clock Speed of 2400 MHz, which refers to the speed at which the computer's memory operates. In this way, data can be transferred at a rate of 2400 million cycles per second, and programs will run smoothly and quickly. The Platform of this motherboard is Windows 10, which means students will be able to install and run Windows 10 on their computers without any compatibility issues. The Memory Storage Capacity is 1GB, and this refers to the amount of random-access memory (RAM) available in the system where the computer can store and access up to 1 gigabyte of data at a time. A socket is where the CPU is connected to the motherboard, and since all motherboards are not the same, they have specific socket types they can support. The ASUS Prime B450M-A/CSM motherboard supports AMD Ryzen processors with AM4 (Advanced Micro Devices Socket AM4) socket type. This socket type facilitates some advanced features. A chipset is responsible for the communication between some of the components of the computer, like the CPU, the memory, and the storage. This motherboard requires an AMD B450 chipset, which has great features, is not expensive, and provides enough connectivity options and performance. This Motherboard has DDR4 memory support for up to 64GB. DDR4 stands for Double Data Rate 4, it offers a faster transfer of data and lowers the power consumption. This Motherboard also has USB 3.1 Gen 2 ports for fast data transfer. Ports are where you plug in things like a headset, keyboard, and USB. Ports allow you to interact with the computer in a certain way. It is also good to have a sort of backup space where you can add new things to increase the computer's performance. That's why we have PCI (Peripheral Component Interconnect) slots that are like reserved spaces in the motherboard where you can plug in new features and new extra cards. PCI slots allow you to upgrade the computer's capabilities by connecting hardware devices to the MB. The ASUS Prime B450M-A/CSM motherboard has PCIe 3.0 slots where you can plug in graphics cards or Wi-Fi adapters. Other features of this motherboard are its fast Connectivity with 1Gb LAN, HDMI 2.0b, D-Sub, and DVI display outputs. The 1Gb LAN (Local Network Area) features mean that your computer is connected to the internet, and there is fast data transfer up to 1 gigabit per second. The HDMI 2.0b (High-Definition Multimedia Interface) is like a link between your computer and your TV or monitor that delivers clear pictures and immersive sound. The D-Sub (D-Subminiature) is a plug to connect your computer to older monitors where you can display your screen. The DVI (Digital Visual Interface) is like a bridge between your computer and your monitor, ensuring great video quality.

Lastly, this motherboard is compatible with a wide range of AMD Ryzen CPUs (1st, 2nd, and 3rd generation AMD Ryzen processors). For the computer that I want to build, my choice of CPU is the Ryzen 3 generation.

Now, let's see why I chose the **AMD Ryzen 3 3200G** CPU. First, it is because of its compatibility with the **ASUS Prime B450M-A/CSM** motherboard, and second, it is for its excellent features. Advanced Micro Devices (AMD) is the manufacturer of this processor. Remember that the chosen motherboard supports AMD AM4 Socket, and the perfect pairing for it is 3rd/2nd/1st Gen AMD Ryzen CPUs. This CPU has a socket type of AM4 and is 2nd Gen, which makes it perfectly compatible with the motherboard. It has 4 Cores, the cache Size is 6MB (L3 Cache), and it has a Speed of 3.6GHz base clock and up to 4.0GHz boost clock. Cores are the individual processing units within a CPU, by having more than one core, the CPU can handle multiple tasks simultaneously. Cache memory is a volatile memory that a CPU uses to store frequently used data. The base clock speed is the frequency at which the CPU operates under normal conditions. It can also boost its clock speed up to 4.0GHz under certain conditions. This CPU would support standard RAM like DDR4. The AMD Ryzen 3 3200G CPU also has other interesting features. It has its own graphics chip (Radeon Vega 8) for gaming and multimedia, you don't need a separate graphics card. Its architecture is efficient since it is designed smartly (12nm Zen+ architecture) to use power wisely while giving a good performance. It only uses about 65 watts, so it lets you save money on electricity bills. With its Advanced Vector Extensions (AVX), it can handle heavy math. Finally, this processor is specifically designed for use in desktop computers (desktop processors). Desktop processors are more powerful than those found in laptops or mobile devices, which is great for students because they allow multitasking to improve performance.

Both the motherboard and CPU are chosen for their affordability, performance, and features. They are compatible with each other, and while they are considered low-end, they still offer good performance for high school tasks such as web browsing, document editing, multimedia consumption, and light gaming. ASUS motherboards are also reliable, which is important for a system intended for educational use where downtime should be minimized.

Motherboard link

CPU link